

MAPPING GENDER TALK ON BLUESKY AHEAD OF THE 2024 U.S. ELECTION

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OVERVIEW

Gender talk on Bluesky just ahead of the last US election: how big, when, and with what tone?

→ Assessing gender politics in the 2024 U.S. race, alongside Bluesky's rapid growth as a post-Musk X alternative

US-politics posts on Bluesky **in the last 2 weeks of Oct 2024**

- Zero-shot BERT to tag gender-related posts (with a dictionary baseline)
- Daily aggregation to see spikes and trends
- Subtopics detected via PMI word co-occurrence network
- RoBERTa sentiment on the tagged subset, applied also to subtopics

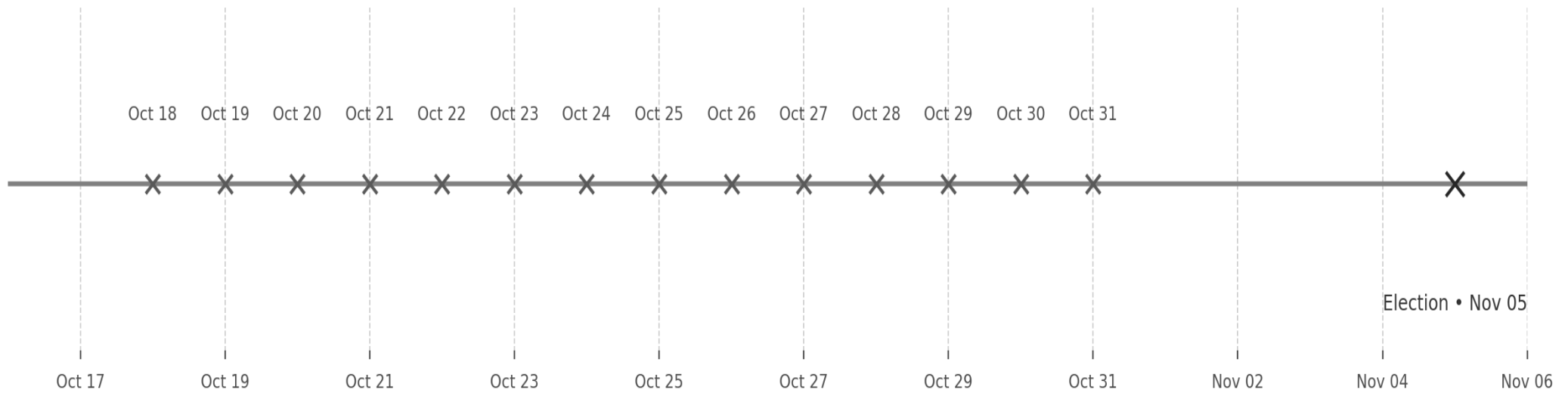
OUR RESEARCH QUESTIONS

Salience over time, event sensitivity, sentiment and subtopic mix

- I. What share of the Bluesky US-politics stream is about gender, in the weeks preceding the elections?
- II. Do salience spikes track campaign moments?
- III. Is the tone around gender mostly negative/neutral/positive?
- IV. Within gender, which subtopic dominates and how does the mix shift?

BLUESKY DATA

- **Source:** Bluesky AppView (AT Protocol) via search_posts
- **Window:** 18 -31 October 2024
- **Queries (US politics):** names + election terms (US election, Republicans, Democrats, ballot, debate, poll...)
- **Fields:** text, createdAt, author handle/DID, like/repost/reply counts, links.
- **Scale:** 58,932 posts (almost 90% of the monthly posts)
- **Limits:** keyword recall; English focus; public posts only.



THE TIMELINE

METHODS

Data prep

- Clean text (normalize whitespace/URLs), drop empty or <15 chars.

Topic flag: gender + daily aggregation + subtopics

- Keyword pre-filter
- Zero-shot MNLI (DistilBERT as compromise for speed & accuracy $\approx 82\%$)
 - Threshold 0.7 $\rightarrow$ binary is_gender_topic

Sentiment (only for gender detected posts)

RoBERTa (twitter-roberta-latest) $\rightarrow$ Positive / Neutral / Negative tone

⌞ _____ ⌟

This strategy allows to give reproducible, label-free estimates of gender salience and tone for the period of interest.

KEYWORDS - PRE-FILTER

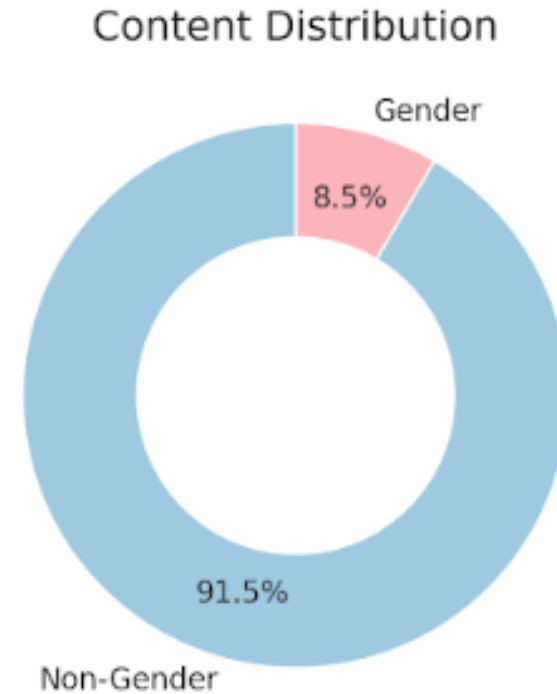
domestic violence
gender discrimination
sexual harassment
workplace equality
women in politics
reproductive rights

woman pay gap
transgender
sexism
misogyny
pro-life
lesbian
maternity leave
female candidate
feminism
misogynist
male
men man
girls
sex
non-binary
gay
boys
pro-choice
sexist
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rape
boy
girl
first woman
queer
sexual assault
feminist
women's rights
paternity leave

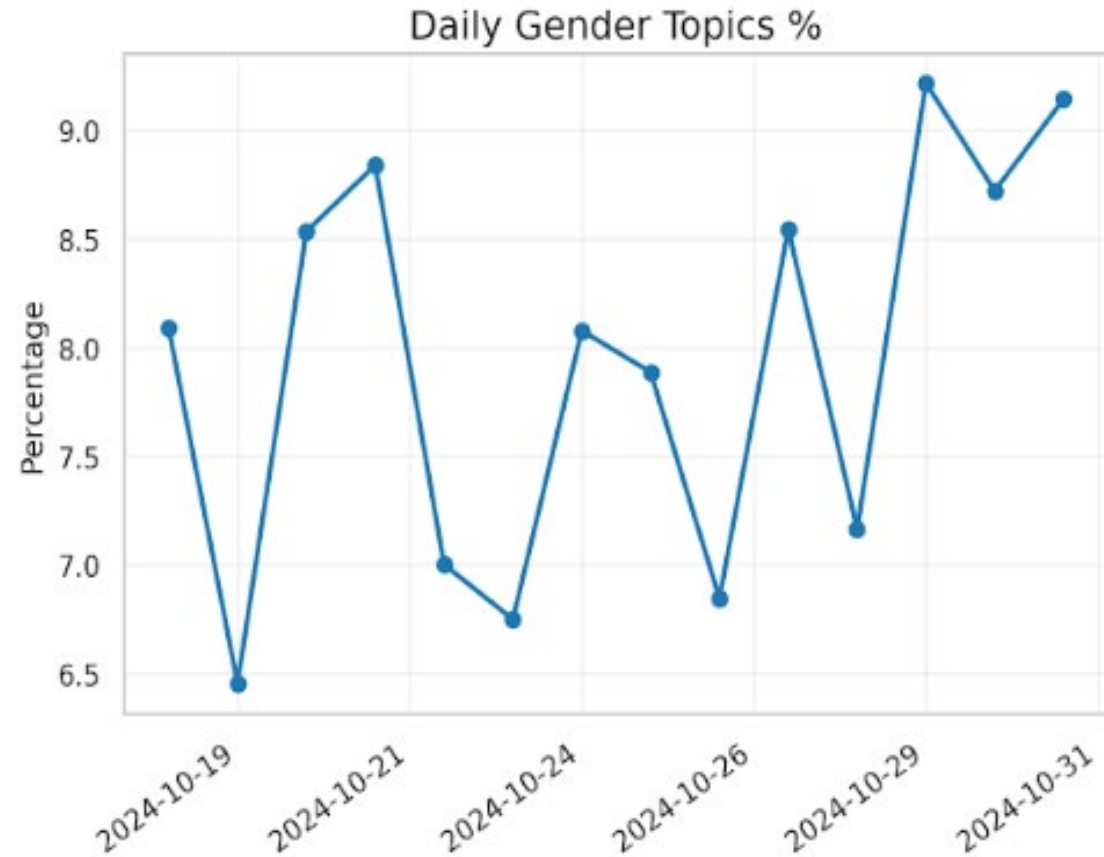
RESULTS: OVERALL SALIENCE

CONTRARY TO
EXPECTATIONS, THE
GENDER TOPIC SHOWED
LIMITED SALIENCE IN
THE CORPUS,
AVERAGING **UNDER 10%**



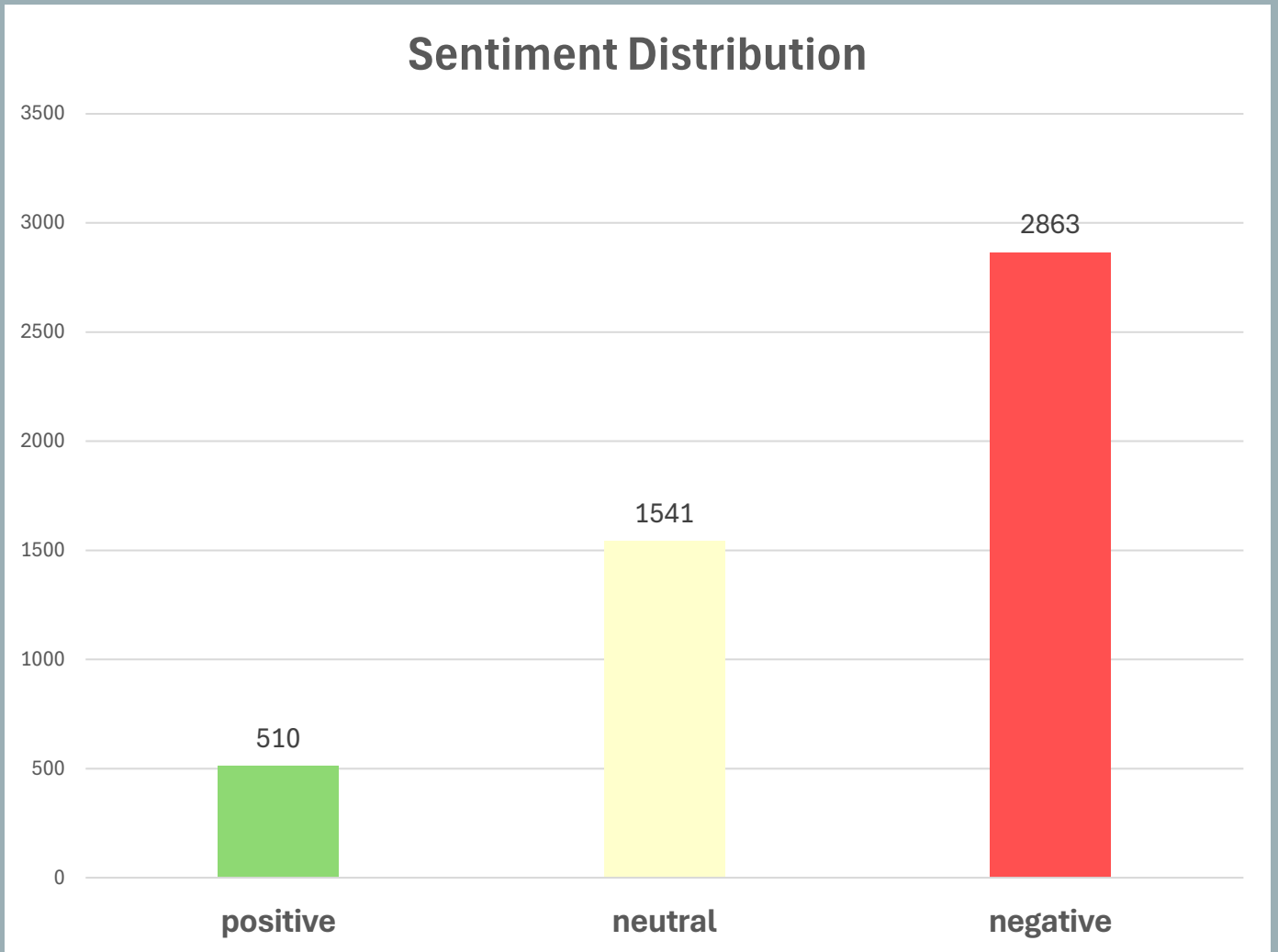
RESULTS: SALIENCE OVER TIME

THE SERIES FLUCTUATES
6–10%, WITH
NOTICEABLE **SPIKES** IN
THE FINAL DAYS



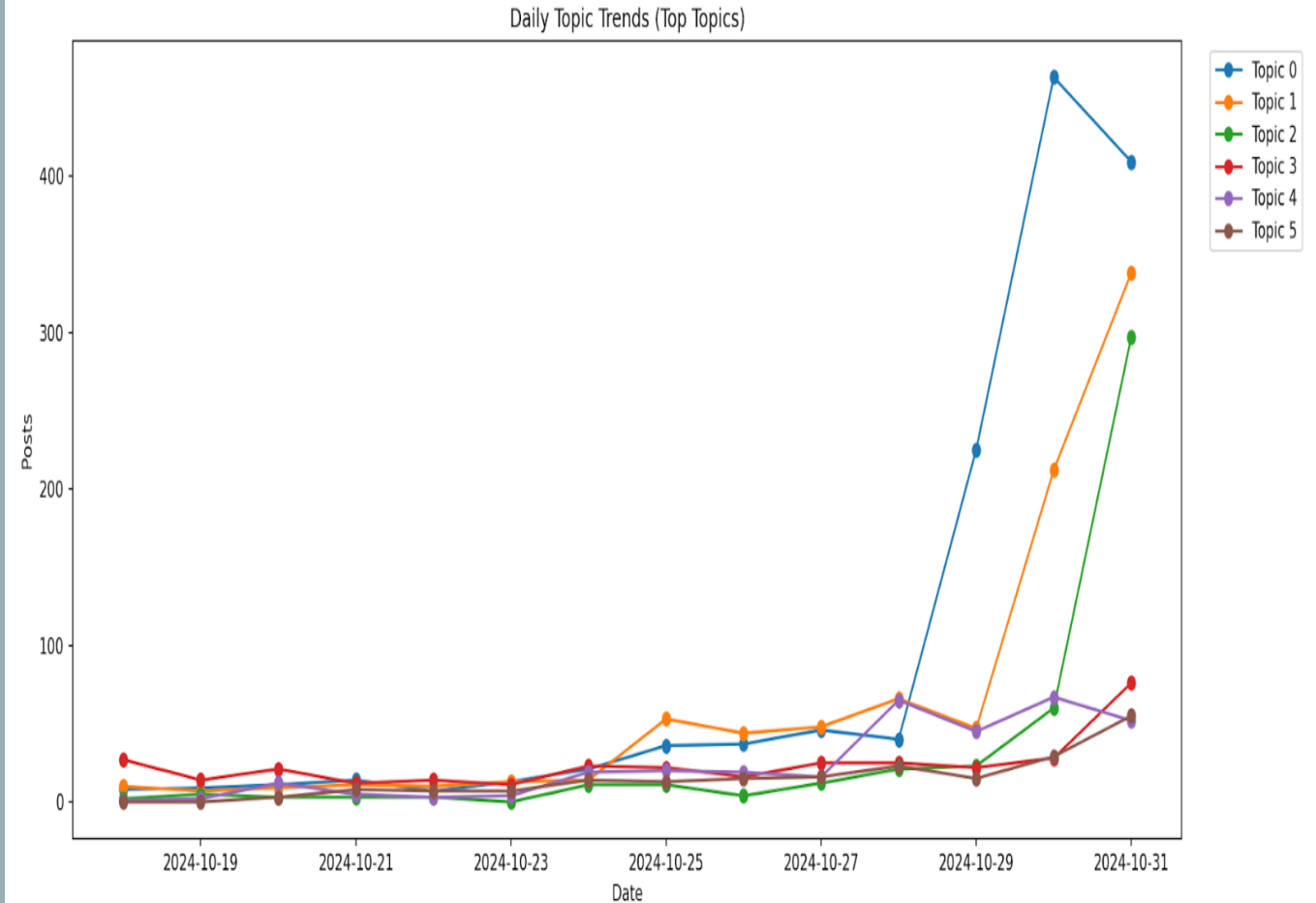
RESULTS: OVERALL SENTIMENT

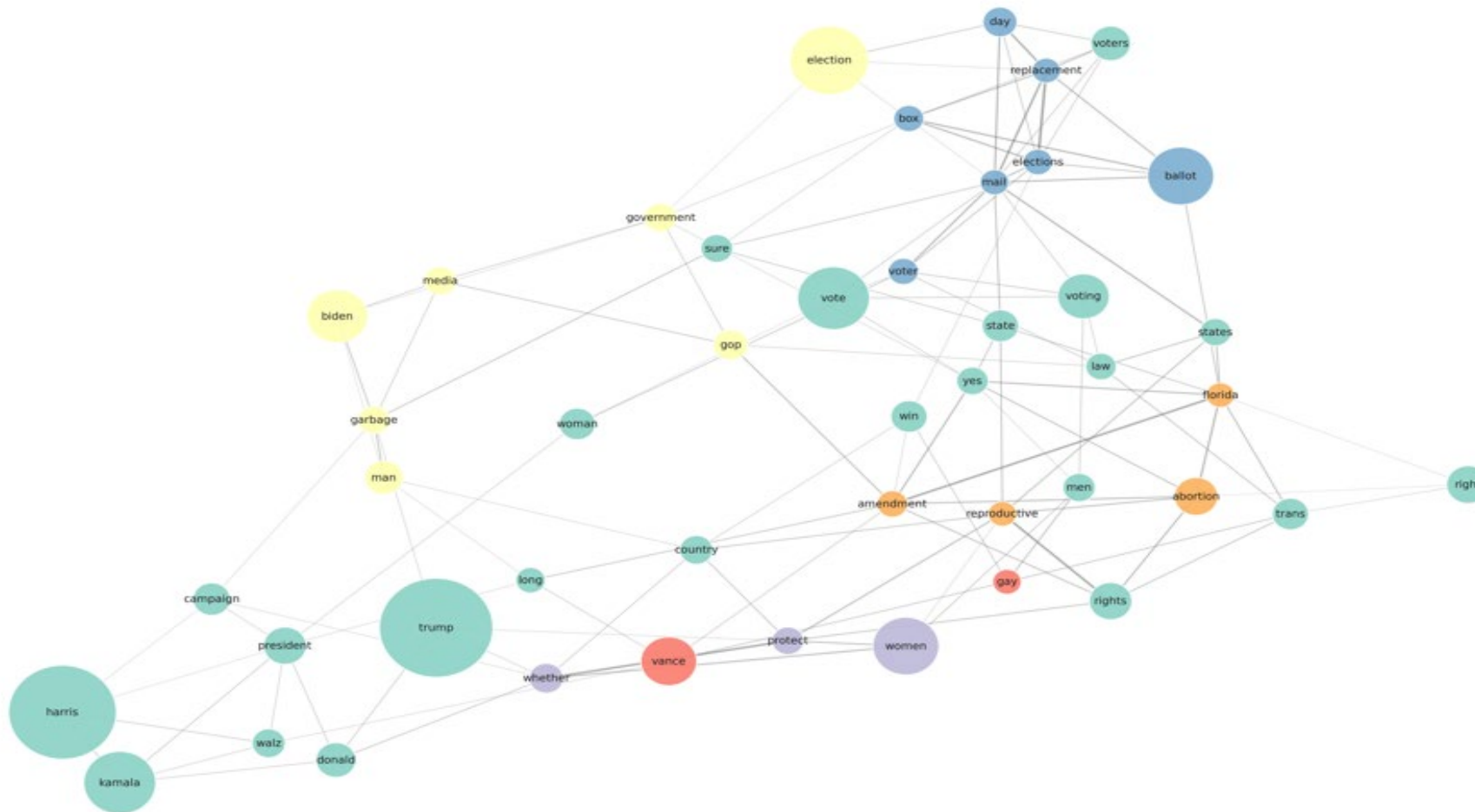
GENDER CONTENT IS
MOSTLY NEGATIVE.
ABOUT 1 IN 10 POSTS
SCORE POSITIVE



RESULTS: SUBTOPICS INSPECTION I

SIX LEADING SUBTOPICS
**WITH MINIMAL
DIFFERENCES, EXCEPT
IN THE LAST DAYS**

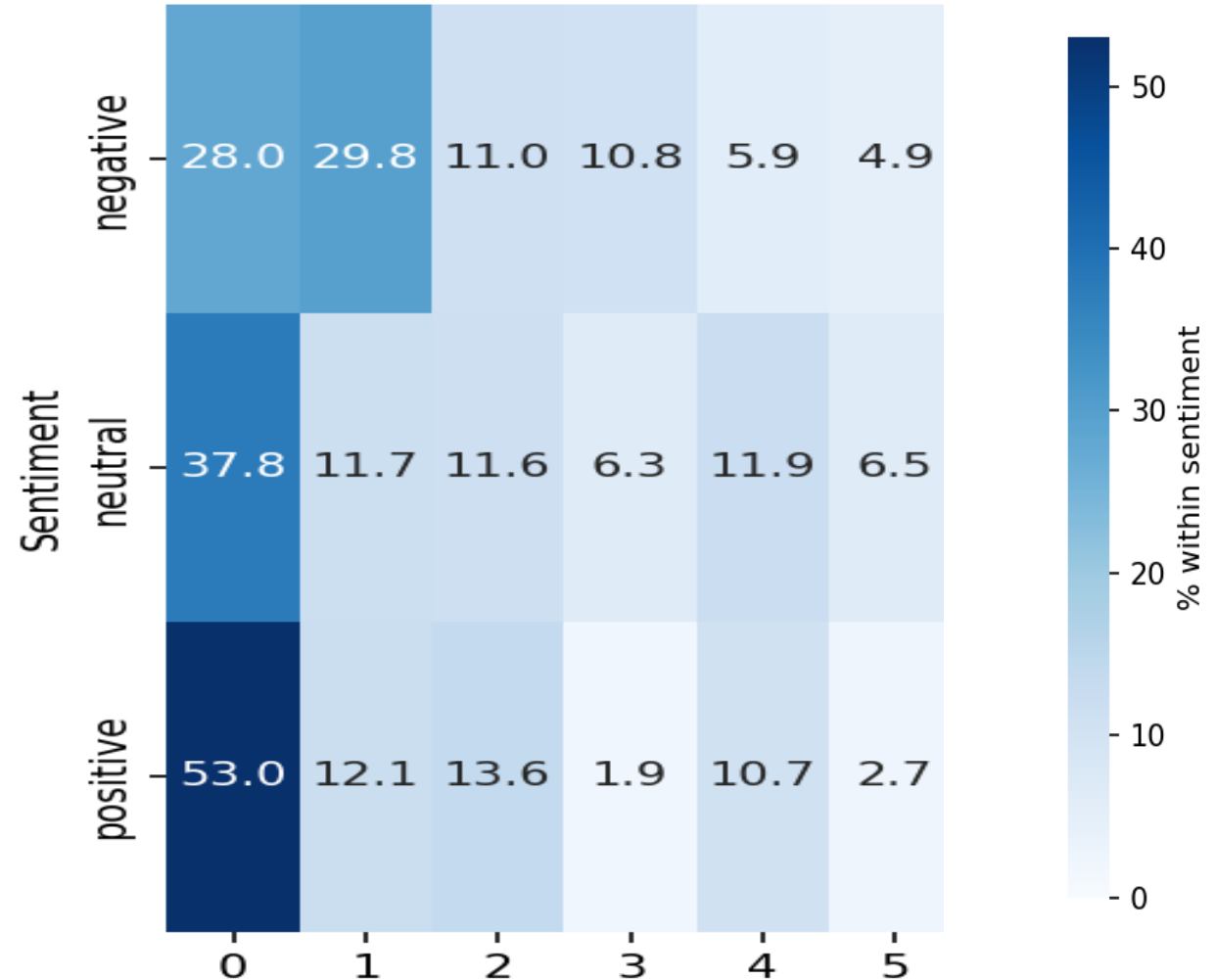




RESULTS: SUBTOPICS INSPECTION II

RESULTS IV – SUBTOPICS HEAT MAP

**SENTIMENT WITHIN
SUBTOPICS 0–5:** 0-2
ARE POSITIVITY-HEAVY,
1–3 CARRY MOST
NEGATIVES, 4 SKEWS
NEUTRAL, 5 RELATIVELY
SMALL.



CONCLUSIONS

- Relatively **low salience overall**: Gender content remained limited—**under 10%** of the corpus on average.
- Stable but **contained variability**: Shares fluctuated within a 6–10% band, with spikes near month-end.
- Most gender-related posts were **negative**; ~10% were positive.
- **Six leading subtopics**: volumes were similar across the top six, diverging only in the final days.
- Most gender references diffuse across the broader political debate; **concrete gender policies** (reproductive rights, abortion) **are comparatively neutral and integrated**, rather than isolated, with other subtopics.

POSSIBLE EXTENSIONS GOING FURTHER

- **Cross-platform replication:** re-run the pipeline on Reddit (r/politics)
- **Widen the time window.** Extend to a 3-month pre-election horizon
- **Parallel case study:** apply the pipeline to Olympic Games (another global event) to compare gender talk with the election sample in August
- **Hate/toxicity audit.** Score the gender subset with a toxicity/hate model
- **Engagement effects.** Analyze likes/reposts and test whether negative vs. positive gender posts receive different engagement
- **Model refinement and fixes.** Improve the subtopics taxonomy

THANK YOU!

